Quantum Field Theory 1 Week 6 Homework

Exercise 6.8

Show that

a)

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho})$$

Clifford algebra: $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$

$$\Rightarrow \mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = \mathrm{Tr}([2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu]\gamma^\rho\gamma^\sigma) = \mathrm{Tr}(2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - \gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma)$$

$$= \mathrm{Tr}(2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - \gamma^\nu[2\eta^{\mu\rho} - \gamma^\rho\gamma^\mu]\gamma^\sigma) = \mathrm{Tr}(2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma + \gamma^\nu\gamma^\rho[2\eta^{\mu\sigma} - \gamma^\sigma\gamma^\mu])$$

$$= 2\eta^{\mu\nu}\underbrace{\mathrm{Tr}(\gamma^\rho\gamma^\sigma)}_{=4\eta^{\rho\sigma}} - 2\eta^{\mu\rho}\underbrace{\mathrm{Tr}(\gamma^\nu\gamma^\sigma)}_{=4\eta^{\nu\sigma}} + 2\eta^{\mu\sigma}\underbrace{\mathrm{Tr}[\gamma^\nu\gamma^\rho]}_{=4\eta^{\nu\rho}} - \underbrace{\mathrm{Tr}(\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu)}_{=\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma)}$$ // cyclic property of Trace

← I'm taking this identity as known since it can be found in book along with it's derivation

$$\Rightarrow 2\,\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 2\cdot4[\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}]$$

$$\Rightarrow \mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}) \quad \square$$

b) $\gamma_\mu \not{a}\not{b}\gamma^\mu = 4a\cdot b$ , $\gamma_\mu\not{a}\not{b}\not{c}\gamma^\mu = -2\not{c}\not{b}\not{a}$ , $\mathrm{Tr}[\gamma_5\not{a}\not{b}] = 0$

Let's start with $\gamma_\mu\not{a}\not{b}\gamma^\mu = 4\cdot a\cdot b$

$$\not{a} = \gamma^\nu a_\nu,\ \not{b} = \gamma^\rho b_\rho \Rightarrow \gamma_\mu\not{a}\not{b}\gamma^\mu = \gamma_\mu\gamma^\nu a_\nu\gamma^\rho b_\rho\gamma^\mu = \gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu a_\nu b_\rho$$

↑ this is true sinc $\gamma_\mu(\dots)\gamma^\mu = \gamma^\mu g_{\mu\nu}(\dots)\gamma^\mu = \gamma^\mu(\dots)g_{\mu\nu}\gamma^\mu = \gamma^\mu(\dots)\gamma_\mu$

Now let's only evaluate $\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu$,

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = (2g^{\mu\nu} - \gamma^\nu\gamma^\mu)\gamma^\rho\gamma_\mu = 2g^{\mu\nu}\gamma^\rho\gamma_\mu - \gamma^\nu(2g^{\mu\rho} - \gamma^\rho\gamma^\mu)\gamma_\mu$$

$$= 2\gamma^\rho\gamma^\nu - 2\gamma^\nu g^{\mu\rho}\gamma_\mu + \gamma^\nu\gamma^\rho\underbrace{\gamma^\mu\gamma_\mu}_{=4} = 2\gamma^\rho\gamma^\nu - 2\gamma^\nu\gamma^\rho + 4\gamma^\nu\gamma^\rho$$

$$\Rightarrow \gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 2\gamma^\rho\gamma^\nu + 2\gamma^\nu\gamma^\rho = 2\{\gamma^\rho,\gamma^\nu\} = 4g^{\nu\rho}$$

$$\Rightarrow \gamma_\mu\not{a}\not{b}\gamma^\mu = 4g^{\nu\rho}a_\nu b_\rho = \underline{\underline{4a\cdot b}} \quad \square$$

$\gamma_\mu \not a \not b \not c \gamma^\mu = \gamma_\mu \gamma^\nu a_\nu \gamma^\rho b_\rho \gamma^\sigma c_\sigma \gamma^\mu = \gamma_\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu a_\nu b_\rho c_\sigma$

$\Rightarrow \gamma_\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu = \gamma_\mu\gamma^\nu\gamma^\rho(2g^{\sigma\mu} - \gamma^\mu\gamma^\sigma) = -\gamma_\mu\gamma^\nu(2g^{\rho\mu}-\gamma^\mu\gamma^\rho)\gamma^\sigma + 2g^{\sigma\mu}\gamma_\mu\gamma^\nu\gamma^\rho$

$= 2\gamma^\sigma\gamma^\nu\gamma^\rho - 2\gamma^\rho\gamma^\nu\gamma^\sigma + \gamma_\mu(2g^{\nu\mu}-\gamma^\mu\gamma^\nu)\gamma^\rho\gamma^\sigma = 2(\gamma^\sigma\gamma^\nu\gamma^\rho - \gamma^\rho\gamma^\nu\gamma^\sigma + \gamma^\nu\gamma^\rho\gamma^\sigma) - \underbrace{\gamma_\mu\gamma^\mu}_{=4}\gamma^\nu\gamma^\rho\gamma^\sigma$

$= 2[\gamma^\sigma\gamma^\nu\gamma^\rho - \gamma^\rho\gamma^\nu\gamma^\sigma - \gamma^\nu\gamma^\rho\gamma^\sigma] = 2[\gamma^\sigma\gamma^\nu\gamma^\rho - \cancel{\gamma^\rho\gamma^\nu\gamma^\sigma} - (2g^{\nu\rho} - \cancel{\gamma^\rho\gamma^\nu})\gamma^\sigma]$

$= 2[\gamma^\sigma\gamma^\nu\gamma^\rho - 2g^{\nu\rho}\gamma^\sigma] = 2\gamma^\sigma\underbrace{[\gamma^\nu\gamma^\rho - 2g^{\nu\rho}]}_{=-\gamma^\rho\gamma^\nu} = -2\gamma^\sigma\gamma^\rho\gamma^\nu$

$\Rightarrow \underline{\underline{\gamma_\mu \not a\not b\not c\gamma^\mu}} = -2\gamma^\sigma\gamma^\rho\gamma^\nu a_\nu b_\rho c_\sigma = \underline{\underline{-2\not c\not b\not a}}$ □

Now $Tr[\gamma_5 \not a \not b] = 0$ :

$\gamma_5 = \gamma^5$

$\gamma^0\gamma^0 = \mathbb{1}$

$\gamma^5\gamma^0 = -\gamma^5\gamma^0$

since $\{\gamma^5, \gamma^\mu\} = 0$

Trace cyclical: $Tr(ABCD) = Tr(BCDA)$ etc.

$Tr[\gamma_5 \not a \not b] = Tr[a_\mu b_\nu \gamma_5 \gamma^\mu \gamma^\nu] = Tr[a_\mu b_\nu \gamma^5 \gamma^0\gamma^0 \gamma^\mu\gamma^\nu]$

$= -Tr[a_\mu b_\nu \gamma^5 \gamma^0 \gamma^\mu \gamma^\nu \gamma^0] = -Tr[a_\mu b_\nu \gamma^5 (\gamma^\mu)^\dagger \underbrace{\gamma^0\gamma^0}_{=\mathbb{1}} (\gamma^\nu)^\dagger]$

$= -Tr[a_\mu b_\nu \gamma^5 (\gamma^\mu)^\dagger (\gamma^\nu)^\dagger] = -Tr[a_\mu b_\nu \gamma^5 \gamma^\nu (\gamma^\mu)^\dagger] = -Tr[\underbrace{a_\mu b_\nu \gamma^5 \gamma^\mu \gamma^\nu}_{\gamma^5 \not a \not b}]$

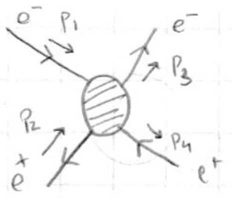
$\Rightarrow Tr[\gamma_5 \not a \not b] = -Tr[\gamma_5 \not a \not b]$

$\rightarrow Tr[\gamma_5 \not a \not b] = 0$

Exercise 6.9 $e^+e^- \rightarrow e^+e^-$ (Babha scattering)

Compute $\left(\frac{d\sigma}{d\cos\theta}\right)$ in the high-energy limit $E_{cm} \gg m_e$

Scattering prosess



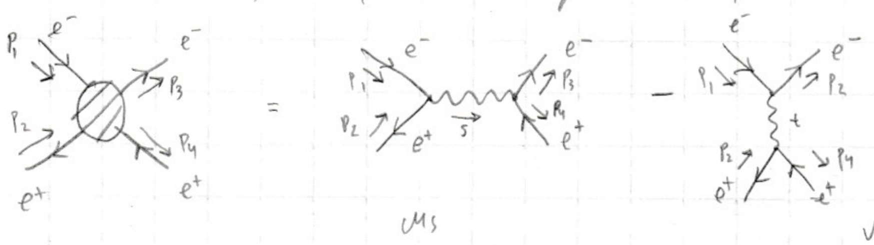

The mandelstam variables are ← $E_{cm} \gg m_e$ limit

$s = (p_1+p_2)^2 = (p_3+p_4)^2 \approx 2p_1p_2 = 2p_3p_4$

$t = (p_1-p_3)^2 = (p_2-p_4)^2 \approx -2p_1p_3 = -2p_2p_4$

$u = (p_1-p_4)^2 = (p_2-p_3)^2 \approx -2p_1p_4 = -2p_2p_3$

The lowest order process for the scattering comes from s-channel & t-channel:

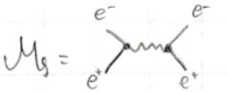



The relative sign difference comes from difference in sign for even and odd number of contraction crossings

We get the expressions for matrix-elements $\mathcal{M}_s$ & $\mathcal{M}_t$ using Feynman-rules for QED

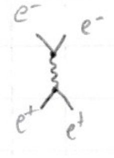
$Q = -1$

Notation: $\bar{v}^r(p_2) = \bar{v}_2^r$

$$\mathcal{M}_s = $$ 


$$= \left[\bar{v}^r(p_2)\,(iQe\gamma^\mu)\,u^s(p_1)\right]\frac{-ig_{\mu\nu}}{s}\left[\bar{u}^{s'}(p_3)\,(iQe\gamma^\mu)\,v^{r'}(p_4)\right]$$

$$\mathcal{M}_s = \frac{ie^2}{s}\left(\bar{v}_2^r\gamma^\mu u_1^s\right)\left(\bar{u}_3^{s'}\gamma_\mu v_4^{r'}\right)$$

$$\mathcal{M}_t = $$ 


$$= \left[\bar{u}^{s'}(p_3)\,(iQe\gamma^\mu)\,u^s(p_1)\right]\frac{-ig_{\mu\nu}}{t}\left[\bar{v}^r(p_2)\,(iQe\gamma^\mu)\,v^{r'}(p_4)\right]$$

$$\mathcal{M}_t = \frac{ie^2}{t}\left(\bar{u}_3^{s'}\gamma^\mu u_1^s\right)\left(\bar{v}_2^r\gamma_\mu v_4^{r'}\right)$$

We can simplify the notation further by noticing that spin indice tells us about the momentum of particle ie : $\bar{u}^{s'} = \bar{u}_3^{s'}$ , $u^s = u_1^s$ , $\bar{v}^r = \bar{v}_2^r$ , $v^{r'} = v_4^{r'}$

So : $\mathcal{M}_s = \frac{ie^2}{s}(\bar{v}^r\gamma^\mu u^s)(\bar{u}^{s'}\gamma_\mu v^{r'})$ and $\mathcal{M}_t = \frac{ie^2}{t}(\bar{u}^{s'}\gamma^\mu u^s)(\bar{v}^r\gamma_\mu v^{r'})$

Since the beam is polarized we must take spin average of the square of the scattering matrix element $\mathcal{M} = \mathcal{M}_s - \mathcal{M}_t$ :

$$\Rightarrow \quad |\mathcal{M}|^2 = \frac{1}{4}\sum_{ss'}\sum_{rr'}|\mathcal{M}_s - \mathcal{M}_t|^2 = \frac{1}{4}\sum_{\substack{ss'\\rr'}}|\mathcal{M}_s|^2 + |\mathcal{M}_t|^2 - 2\mathrm{Re}\{\mathcal{M}_s^\dagger\mathcal{M}_t\}$$

Let's compute $|M|^2$ term by term :

$$\langle|\mathcal{M}_s|^2\rangle = \frac{1}{4}\sum_{\substack{ss'\\rr'}}|\mathcal{M}_s|^2 = \frac{1}{4}\sum_{\substack{ss'\\rr'}}\frac{e^4}{s^2}(\bar{v}^r\gamma^\mu u^s)(\bar{u}^{s'}\gamma_\mu v^{r'})\left[(\bar{v}^r\gamma^\nu u^s)(\bar{u}^{s'}\gamma_\nu v^{r'})\right]^\dagger$$

$$= \frac{e^4}{s^2 4}\sum_{\substack{ss'\\rr'}}(\bar{v}^r\gamma^\mu u^s)(\bar{u}^{s'}\gamma_\mu v^{r'})(\bar{u}^s\gamma^\nu v^r)(\bar{v}^{r'}\gamma_\nu u^{s'})$$

We can write the Lorentz vectors in component form:

$$(\bar{v}^r\gamma^\mu u^s) = \sum_{ab}\bar{v}_a^r\gamma_{ab}^\mu u_b^s \equiv \bar{v}_a^r\gamma_{ab}^\mu u_b^s$$

$$\langle|\mathcal{M}_s|^2\rangle = \frac{e^4}{4s^2}\sum_{\substack{ss'\\rr'}}(\bar{v}_a^r\gamma_{ab}^\mu u_b^s)(\bar{u}_c^{s'}\gamma_{\mu cd}v_d^{r'})(\bar{u}_e^s\gamma_{ef}^\nu v_f^r)(\bar{v}_g^{r'}\gamma_{\nu gh}u_h^{s'})$$

$$= \frac{e^4}{4s^2}\gamma_{ab}^\mu\Big(\sum_s u_b^s\bar{u}_e^s\Big)\gamma_{ef}^\nu\Big(\sum_r v_f^r\bar{v}_a^r\Big)\gamma_{\mu cd}\Big(\sum_{r'}v_d^{r'}\bar{v}_g^{r'}\Big)\gamma_{\nu gh}\Big(\sum_{s'}u_h^{s'}\bar{u}_c^{s'}\Big)$$

Now we once again write the momentum information as indices since we will sum over spin indices using spin completeness relations.

Spin completeness relation $\sum_s u^s(p_1)\bar{u}^s(p_1) = \not{p}_1 + m$ , $\sum_s v^s(p)\bar{v}^s(p) = \not{p} - m$

We are in relativistic limit of $E_{cm} \gg m_e \Rightarrow p_1, p_2, p_3, p_4 \gg m_e \Rightarrow \underline{\sum u^s(p)\bar{u}^s(p) \simeq \not{p}}$

$$\Rightarrow \quad \langle|\mathcal{M}_s|^2\rangle = \frac{e^4}{4s^2}\underbrace{\left(\gamma_{ab}^\mu\not{p}_{1\,be}\gamma_{ef}^\nu\not{p}_{2\,fa}\right)}_{\text{this is sum of diagonal component of } \gamma^\mu\not{p}_1\gamma^\nu\not{p}_2 \text{ which is just a Trace}}\underbrace{\left(\gamma_{\mu cd}\not{p}_{4\,dg}\gamma_{\nu gh}\not{p}_{3\,hc}\right)}_{=(\gamma_\mu\not{p}_4\gamma_\nu\not{p}_3)_{cc} = \mathrm{Tr}(\gamma_\mu\not{p}_4\gamma_\nu\not{p}_3)}$$

$$\Rightarrow \langle|M_s|^2\rangle = \frac{e^4}{4s^2}\,\mathrm{Tr}\left(\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_2\right)\mathrm{Tr}\left(\gamma_\mu \not{p}_4 \gamma_\nu \not{p}_3\right) \qquad // \quad \not{p} = \gamma^\varepsilon p_\varepsilon$$

$$= \frac{e^4}{4s^2}\,\mathrm{Tr}\left(\gamma^\mu \gamma^\rho p^1_\rho \gamma^\nu \gamma^\sigma p^2_\sigma\right)\mathrm{Tr}\left(g_{\mu\lambda}\gamma^\lambda \gamma^\delta p^4_\delta\, g_{\nu\kappa}\gamma^\kappa \gamma^\varepsilon p^3_\varepsilon\right)$$

$$= \frac{e^4}{4s^2}\, p^1_\rho p^2_\sigma\, \mathrm{Tr}\left(\gamma^\mu\gamma^\rho\gamma^\nu\gamma^\sigma\right) g_{\mu\lambda}g_{\nu\kappa}\, p^4_\delta p^3_\varepsilon\, \mathrm{Tr}\left(\gamma^\lambda\gamma^\delta\gamma^\kappa\gamma^\varepsilon\right) \qquad // \quad p^1 \equiv p_1 \text{ here for easier indexing}$$

Now we use the trace identity shown in previous exercise:

$$\mathrm{Tr}\left(\gamma^\mu\gamma^\rho\gamma^\nu\gamma^\sigma\right) = 4\left(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\nu}g^{\rho\sigma} + g^{\mu\sigma}g^{\rho\nu}\right)$$

$$\Rightarrow \langle|M_s|^2\rangle = \frac{e^4}{4s^2}\, 4 p^1_\rho p^2_\sigma \left(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\nu}g^{\rho\sigma} + g^{\mu\sigma}g^{\rho\nu}\right) 4 p^4_\delta p^3_\varepsilon\, g_{\mu\lambda}g_{\nu\kappa}\left(g^{\lambda\delta}g^{\kappa\varepsilon} - g^{\lambda\kappa}g^{\delta\varepsilon} + g^{\lambda\varepsilon}g^{\delta\kappa}\right)$$

$$= 4\frac{e^4}{s^2}\left(p_1^\mu p_2^\nu - g^{\mu\nu}(p_1\cdot p_2) + p_1^\nu p_2^\mu\right) p^4_\delta p^3_\varepsilon \left(\delta_\mu^{\ \delta}\delta_\nu^{\ \varepsilon} - \underbrace{g_{\mu\lambda}\delta_\nu^{\ \lambda}}_{g_{\mu\nu}} g^{\delta\varepsilon} + \delta_\mu^{\ \varepsilon}\delta_\nu^{\ \delta}\right)$$

$$= 4\frac{e^4}{s^2}\left(p_1^\mu p_2^\nu - g^{\mu\nu}(p_1\cdot p_2) + p_1^\nu p_2^\mu\right)\left(p^4_\mu p^3_\nu - g_{\mu\nu}(p_4\cdot p_3) + p^4_\nu p^3_\mu\right)$$

$$= 4\frac{e^4}{s^2}\Big((p_1\cdot p_4)(p_2\cdot p_3) - \cancel{(p_1\cdot p_2)(p_4\cdot p_3)} + (p_1\cdot p_3)(p_2\cdot p_4) - \cancel{(p_1\cdot p_2)(p_4\cdot p_3)} + \underbrace{g^{\mu\nu}g_{\mu\nu}}_{=4}\cancel{(p_1\cdot p_2)(p_4\cdot p_3)}$$
$$- \cancel{(p_1\cdot p_2)(p_4\cdot p_3)} + (p_1\cdot p_3)(p_2\cdot p_4) - \cancel{(p_1\cdot p_2)(p_4\cdot p_3)} + (p_1\cdot p_4)(p_2\cdot p_3)\Big)$$

$$= \frac{e^4}{s^2}\,4\cdot 2\left((p_1\cdot p_4)(p_2\cdot p_3) + (p_1\cdot p_3)(p_2\cdot p_4)\right) \qquad // \text{ Mandelstam: } u = -2p_1\cdot p_4 = -2p_2\cdot p_3$$
$$t = -2p_1\cdot p_3 = -2p_2\cdot p_4$$

$$= \frac{e^4}{s^2}\,4\cdot 2\left(\frac{1}{4}u^2 + \frac{1}{4}t^2\right) = \frac{2e^4}{s^2}(u^2+t^2)$$

$$\Rightarrow \langle|M_s|^2\rangle = \frac{2e^4}{s^2}(u^2+t^2)$$

Now we calculate $\langle|M_t|^2\rangle = \frac{1}{4}\sum_{\substack{ss'\\rr'}}|M_t|^2$

$$\langle|M_t|^2\rangle = \frac{1}{4}\sum_{\substack{ss'\\rr'}}\frac{e^4}{t^2}\left(\bar{u}^{s'}\gamma^\mu u^s\right)\left(\bar{v}^r\gamma_\mu v^{r'}\right)\left(\bar{u}^s\gamma^\nu u^{s'}\right)\left(\bar{v}^{r'}\gamma_\nu v^r\right) \qquad // \text{ We repeat same steps as we did for } \langle|M_s|^2\rangle$$

$$= \frac{e^4}{4t^2}\sum_{\substack{ss'\\rr'}} \bar{u}^{s'}_a \gamma^\mu_{ab} u^s_b\, \bar{v}^r_c \gamma_{\mu cd} v^{r'}_d\, \bar{u}^s_e \gamma^\nu_{ef} u^{s'}_f\, \bar{v}^{r'}_g \gamma_{\nu gh} v^r_h$$

$$= \frac{e^4}{4t^2}\left[\gamma^\mu_{ab}\left(\sum_s u^s_b \bar{u}^s_e\right)\gamma^\nu_{ef}\left(\sum_{s'} u^{s'}_f \bar{u}^{s'}_a\right)\right]\left[\gamma_{\mu cd}\left(\sum_{r'} v^{r'}_d \bar{v}^{r'}_g\right)\gamma_{\nu gh}\left(\sum_r v^r_h \bar{v}^r_c\right)\right]$$

$$= \frac{e^4}{4t^2}\,\mathrm{Tr}\left[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_3\right]\mathrm{Tr}\left[\gamma_\mu \not{p}_4 \gamma_\nu \not{p}_2\right]$$

From $\langle|M_s|^2\rangle$ we got $\frac{e^4}{4s^2}\mathrm{Tr}\left[\gamma^\mu \not{p}_1\gamma^\nu \not{p}_2\right]\mathrm{Tr}\left[\gamma_\mu \not{p}_4 \gamma_\nu \not{p}_3\right] = \frac{e^4}{s^2}4\cdot 2\left((p_1\cdot p_4)(p_2\cdot p_3) + (p_1\cdot p_3)(p_2\cdot p_4)\right)$

By just changing indices of $\not{p}$ we get result for $\langle|M_t|^2\rangle$

$$\Rightarrow \frac{e^4}{4t^2}\mathrm{Tr}\left[\gamma^\mu \not{p}_1\gamma^\nu \not{p}_3\right]\mathrm{Tr}\left[\gamma_\mu \not{p}_4\gamma_\nu \not{p}_2\right] = \frac{e^4}{t^2}\,4\cdot 2\left(\underbrace{(p_1\cdot p_4)}_{-\frac{1}{2}u}\underbrace{(p_3\cdot p_2)}_{-\frac{1}{2}u} + \underbrace{(p_1\cdot p_2)}_{\frac{1}{2}s}\underbrace{(p_3\cdot p_4)}_{\frac{1}{2}s}\right)$$

$$\Rightarrow \langle|M_t|^2\rangle = \frac{2e^4}{t^2}(u^2+s^2)$$

Now we calculate $\langle M_{st} \rangle \equiv \frac{1}{4} \sum_{\substack{ss' \\ rr'}} -2\, \mathrm{Re}\{ M_s^{\dagger} M_t \}$

$$\langle M_{st} \rangle = -\frac{1}{2} \sum_{\substack{ss' \\ rr'}} \mathrm{Re}\left\{ -\frac{ie^2}{s} (\bar{u}^s \gamma^\mu v^r)(\bar{v}^{r'} \gamma_\mu u^{s'}) \frac{ie^2}{t} (\bar{u}^{s'} \gamma^\nu u^s)(\bar{v}^r \gamma_\nu v^{r'}) \right\}$$

$$= -\frac{e^4}{2st} \sum_{\substack{ss' \\ rr'}} \mathrm{Re}\{ \bar{u}^s_a \gamma^\mu_{ab} v^r_b \, \bar{v}^{r'}_c \gamma_{\mu cd} u^{s'}_d \, \bar{u}^{s'}_e \gamma^\nu_{ef} u^s_f \, \bar{v}^r_g \gamma_{\nu gh} v^{r'}_h \}$$

$$= -\frac{1}{2} \frac{e^4}{st} \mathrm{Re}\left\{ \gamma^\mu_{ab} \Big( \sum_r v^r_b \bar{v}^r_g \Big) \gamma_{\nu gh} \Big( \sum_{r'} v^{r'}_h \bar{v}^{r'}_c \Big) \gamma_{\mu cd} \Big( \sum_{s'} u^{s'}_d \bar{u}^{s'}_e \Big) \gamma^\nu_{ef} \Big( \sum_s u^s_f \bar{u}^s_a \Big) \right\}$$

$$= -\frac{1}{2} \frac{e^4}{st} \mathrm{Re}\left\{ Tr\left[ \gamma^\mu \not{p}_2 \gamma_\nu \not{p}_4 \gamma_\mu \not{p}_3 \gamma^\nu \not{p}_1 \right] \right\}$$

$$= -\frac{1}{2} \frac{e^4}{st} \mathrm{Re}\left\{ Tr\left[ g_{\nu\rho} \gamma^\mu \not{p}_2 \gamma^\rho \not{p}_4 \gamma_\mu \not{p}_3 \gamma^\nu \not{p}_1 \right] \right\} \quad \Big\| \quad \not{p} = \gamma^\alpha p_\alpha$$

$$= -\frac{e^4}{2st} \mathrm{Re}\{ g_{\nu\rho} \, p_{2\alpha} p_{4\beta} p_{3\delta} p_{1\varepsilon} \, Tr[ \gamma^\mu \gamma^\alpha \gamma^\rho \gamma^\beta \gamma_\mu \gamma^\delta \gamma^\nu \gamma^\varepsilon ] \}$$

We now use relation $\gamma^\mu \gamma^\alpha \gamma^\rho \gamma^\beta \gamma_\mu = -2 \gamma^\beta \gamma^\rho \gamma^\alpha$

$$\langle M_{st} \rangle = -\frac{e^4}{2st} \mathrm{Re}\{ p_{2\alpha} p_{4\beta} p_{3\delta} p_{1\varepsilon} \, Tr[ -2 \gamma^\beta \gamma^\nu \gamma^\alpha \gamma^\delta \gamma_\nu \gamma^\varepsilon ] \}$$

|| $g_{\nu\rho} \gamma^\rho \gamma_\nu = \gamma_\nu \gamma^\nu = \gamma^\nu \gamma_\nu$
Now we use $\gamma^\nu \gamma^\alpha \gamma^\delta \gamma_\nu = 4 g^{\alpha\delta}$

$$= -\frac{e^4}{2st} \mathrm{Re}\{ -2 \, p_{2\alpha} p_{4\beta} p_{3\delta} p_{1\varepsilon} \underbrace{Tr[\gamma^\beta \gamma^\varepsilon]}_{4 g^{\beta\varepsilon}} g^{\alpha\delta} \cdot 4 \} \quad \Big\| \quad Tr(\gamma^\mu \gamma^\nu) = 4 g^{\mu\nu}$$

$$= 4 \frac{e^4}{st} \mathrm{Re}\{ p_{2\alpha} p_{4\beta} p_{3\delta} p_{1\varepsilon} \cdot 4 g^{\beta\varepsilon} g^{\alpha\delta} \}$$

$$= 16 \frac{e^4}{st} \mathrm{Re}\left( (p_2 \cdot p_3)(p_4 \cdot p_1) \right) \quad \Big\| \quad (p_2 \cdot p_3) = -\frac{1}{2} u = (p_4 \cdot p_1)$$

$\langle M_{st} \rangle = 4 \frac{e^4}{st} u^2$ , Finally we can now get $|M|^2 = \frac{1}{4} \sum_{ss'} |M_s - M_t|^2$

$$\Rightarrow |M|^2 = \langle |M_s|^2 \rangle + \langle |M_t|^2 \rangle + \langle M_{st} \rangle$$

$$= \frac{2e^4}{s^2} (u^2 + t^2) + \frac{2e^4}{t^2} (u^2 + s^2) + 4 \frac{e^4}{st} u^2$$

$$= 2e^4 \left[ \frac{u^2 + t^2}{s^2} + \frac{u^2 + s^2}{t^2} + 2 \frac{u^2}{st} \right]$$

The differential cross-section in Centre of Mass frame is

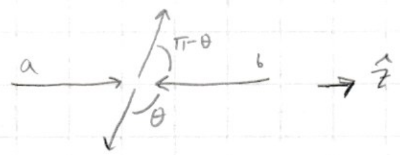
$$(d\sigma / d\Omega)_{CM} = \frac{1}{F} \frac{|p_1|}{16 \pi^2 E_{cm}} |M|^2$$

|| Relativistic limit: $|p_1| = E = \frac{1}{2}\sqrt{s} = \frac{1}{2\sqrt{s}} \lambda^{1/2}(s,0,0)$
$F = 2 \lambda^{1/2}(s,0,0)$ , $E_{cm} = \sqrt{s}$

$$\Rightarrow \left(\frac{d\sigma}{d\Omega}\right)_{CM} = \frac{1}{64\pi^2 s}|\mathcal{M}|^2 \quad \Bigg\| \quad \frac{d\sigma}{d\Omega} = \frac{d\sigma}{d\phi\, d\theta\,(\sin\theta)} \quad \Bigg\| \quad \frac{d\cos\theta}{d\theta} = -\sin\theta$$

$$\Rightarrow \frac{1}{d\theta \sin\theta} = -\frac{1}{d\cos\theta}$$

$|\mathcal{M}|^2$ doesn't have $\phi$-dependency since only $\theta$ is enough to describe a 2-to-2 scattering prosess. So there will always be coordinate system where particles only move along a plane defined by $z$-axis and $\theta$-angle. So we can integrate over $\phi$:




$$\left(\frac{d\sigma}{d\cos\theta}\right)_{CM} = -\int_0^{2\pi} d\phi \frac{1}{64\pi^2 s}|\mathcal{M}|^2 = -\frac{1}{32\pi s}|\mathcal{M}|^2$$

$$\left(\frac{d\sigma}{d\cos\theta}\right)_{CM} = -\frac{2e^4}{32\pi s}\left[\frac{u^2+t^2}{s^2} + \frac{u^2+s^2}{t^2} + \frac{2u^2}{st}\right] \quad \Bigg\| \quad \alpha = \frac{e^2}{4\pi} \Rightarrow e^4 = \alpha^2(4\pi)^2$$

$$= -\frac{\alpha^2(4\pi)^2}{16\pi s}\left[\frac{u^2+t^2}{s^2} + \frac{u^2+s^2}{t^2} + \frac{2u^2}{st}\right]$$

$$\frac{d\sigma}{d\cos\theta} = -\frac{\alpha^2\pi}{s}\left[\frac{u^2+t^2}{s^2} + \frac{u^2+s^2}{t^2} + \frac{2u^2}{st}\right]$$

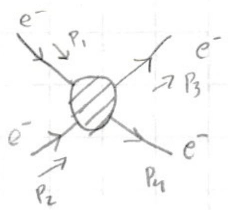
$$\Rightarrow \boxed{\frac{d\sigma}{d\cos\theta} = -\frac{\alpha^2\pi}{s}\left[\frac{u^2+t^2}{s^2} + \frac{u^2+s^2}{t^2} + \frac{2u^2}{st}\right]}$$

We defined this in terms of CM-frame but it is valid in all frames since it only depends on Lorentz invariant Mandestam variables.
If we want we can make the expression positive via coordinate transformation such that $d\theta' \sin(\theta') = d\theta \sin(\theta)$
$\Rightarrow d\theta'\sin(\theta') = d\cos\theta$, but I will not be doing that

## Exercise 6.10

$e^-e^- \to e^-e^-$ scattering in high-energy (relativistic) limit $E_{cm} \gg m_e$

Scattering process:

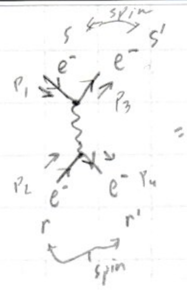



The mandelstam variables are (in $E_{cm} \gg m_e$ limit):
$s \approx 2P_1P_2 = 2P_3P_4$, $t \approx -2P_1P_3 = -2P_2P_4$
$u \approx -2P_1P_4 = -2P_2P_3$

The lowest order process for the scattering comes from t-channel & u-channel


The relative sign difference comes from different numbers of contraction crossings.

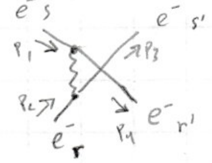
We now use the Feynman rules for QED to get the matrix elements $\mathcal{M}_t$ & $\mathcal{M}_u$.



$$\mathcal{M}_t = \quad = \left[\bar{u}^{s'}(p_3)\,(iQe\gamma^\mu)\,u^s(p_1)\right]\frac{-ig_{\mu\nu}}{t}\left[\bar{u}^{r'}(p_4)\,(iQe\gamma^\mu)\,u^r(p_2)\right]$$

$$= \frac{ie^2}{t}\left(\bar{u}^{s'}\gamma^\mu u^s\right)\left(\bar{u}^{r'}\gamma_\mu u^r\right)$$

We drop the momentum labels since we know momentum from spin indices since they have 1-to-1 correspondence



$$\mathcal{M}_u = \quad = \left[\bar{u}^{r'}(p_4)\,(iQe\gamma^\mu)\,u^s(p_1)\right]\frac{-ig_{\mu\nu}}{u}\left[\bar{u}^{s'}(p_3)(iQe\gamma^\mu)u^r(p_2)\right]$$

$$= \frac{ie^2}{u}\left(\bar{u}^{r'}\gamma^\mu u^s\right)\left(\bar{u}^{s'}\gamma_\mu u^r\right)$$

The beam is unpolarized: Need to take spin-average:

$$|\mathcal{M}|^2 = \frac{1}{4}\sum_{\substack{rr'\\ss'}}|\mathcal{M}_t - \mathcal{M}_u|^2 = \frac{1}{4}\sum_{\substack{ss'\\rr'}}|\mathcal{M}_t|^2 + |\mathcal{M}_u|^2 - 2\,\mathrm{Re}\{\mathcal{M}_t^\dagger \mathcal{M}_u\}$$

Let's calculate term by term:

$$\langle|\mathcal{M}_t|^2\rangle = \frac{1}{4}\sum_{\substack{ss'\\rr'}}|\mathcal{M}_t|^2 = \frac{1}{4}\sum_{\substack{ss'\\rr'}}\frac{e^4}{t^2}\left(\bar{u}^{s'}\gamma^\mu u^s\right)\left(\bar{u}^{r'}\gamma_\mu u^r\right)\left[\left(\bar{u}^{s'}\gamma^\nu u^s\right)\left(\bar{u}^{r'}\gamma_\nu u^r\right)\right]^\dagger$$

$$= \frac{1}{4}\frac{e^4}{t^2}\sum_{ss'rr'}\left(\bar{u}^{s'}\gamma^\mu u^s\right)\left(\bar{u}^{r'}\gamma_\mu u^r\right)\left(\bar{u}^{s}\gamma^\nu u^{s'}\right)\left(\bar{u}^{r}\gamma_\nu u^{r'}\right)$$

Let's write this in component form: $(\bar{u}^{s'}\gamma^\mu u^s) = \sum_{ab}\bar{u}^{s'}_a\gamma^\mu_{ab}u^s_b \equiv \bar{u}^{s'}_a\gamma^\mu_{ab}u^s_b$

$$\Rightarrow \langle|\mathcal{M}_t|^2\rangle = \frac{1}{4}\frac{e^4}{t^2}\sum_{ss'rr'}\bar{u}^{s'}_a\gamma^\mu_{ab}u^s_b\,\bar{u}^{r'}_c\gamma_{\mu cd}u^r_d\,\bar{u}^s_e\gamma^\nu_{ef}u^{s'}_f\,\bar{u}^r_g\gamma_{\nu gh}u^{r'}_h$$

$$= \frac{1}{4}\frac{e^4}{t^2}\underbrace{\gamma^\mu_{ab}\Big(\sum_s u^s_b\bar{u}^s_e\Big)\gamma^\nu_{ef}\Big(\sum_{s'}u^{s'}_f\bar{u}^{s'}_a\Big)}_{=\,\mathrm{Tr}(\gamma^\mu \not{p}_1\gamma^\nu\not{p}_3)}\underbrace{\gamma_{\mu cd}\Big(\sum_r u^r_d\bar{u}^r_g\Big)\gamma_{\nu gh}\Big(\sum_{r'}u^{r'}_h\bar{u}^{r'}_c\Big)}_{=\,\mathrm{Tr}[\gamma_\mu\not{p}_2\gamma_\nu\not{p}_4]}$$

high-energy limit $\sum_s u^s\bar{u}^s = \not{p}$

$$= \frac{1}{4}\frac{e^4}{t^2}\mathrm{Tr}\left[\gamma^\mu\not{p}_1\gamma^\nu\not{p}_3\right]\mathrm{Tr}\left[\gamma_\mu\not{p}_2\gamma_\nu\not{p}_4\right]$$

We have solved these traces in previous exercise so let us use them:

$$\frac{e^4}{4t^2}\mathrm{Tr}\left[\gamma^\mu\not{p}_a\gamma^\nu\not{p}_b\right]\mathrm{Tr}\left[\gamma_\mu\not{p}_c\gamma_\nu\not{p}_d\right] = 8\frac{e^4}{t^2}\Big((p_a\cdot p_c)(p_b\cdot p_d) + (p_a\cdot p_d)(p_b\cdot p_c)\Big)$$

This gives us $\langle|\mathcal{M}_t|^2\rangle = 8\frac{e^4}{t^2}\left[(p_1\cdot p_2)(p_3\cdot p_4) + (p_1\cdot p_4)(p_3\cdot p_2)\right]$

$(p_1\cdot p_2) = \frac{1}{2}s = (p_3\cdot p_4)$

$(p_1\cdot p_4) = -\frac{1}{2}u = (p_3\cdot p_2)$

$$\langle|\mathcal{M}_t|^2\rangle = \frac{8e^4}{t^2}\left[s^2 + u^2\right]\frac{1}{4}$$

$$\langle|\mathcal{M}_t|^2\rangle = \frac{2e^4}{t^2}\left(u^2 + s^2\right)$$

$$\langle|\mathcal{M}_u|^2\rangle = \frac{1}{4}\sum_{ss'rr'} |\mathcal{M}_u|^2 = \frac{1}{4}\sum_{ss'rr'} \frac{e^4}{u^2} (\bar{u}^{r'}\gamma^\mu u^s)(\bar{u}^{s'}\gamma_\mu u^r)\left[(\bar{u}^{r'}\gamma^\nu u^s)(\bar{u}^{s'}\gamma_\nu u^r)\right]^\dagger$$

In component form: $\langle|\mathcal{M}_u|^2\rangle = \frac{e^4}{4u^2}\sum_{ss'rr'} \bar{u}^{r'}_a \gamma^\mu_{ab} u^s_b \bar{u}^{s'}_c \gamma_{\mu cd} u^r_d \bar{u}^s_e \gamma^\nu_{ef} u^{r'}_f \bar{u}^r_g \gamma_{\nu gh} u^{s'}_h$

$$\langle|\mathcal{M}_u|^2\rangle = \frac{e^4}{4u^2} \underbrace{\gamma^\mu_{ab}\Big(\sum_s u^s_b \bar{u}^s_e\Big)\gamma^\nu_{ef}\Big(\sum_{r'} u^{r'}_f \bar{u}^{r'}_a\Big)}_{=\mathrm{Tr}(\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_4)} \underbrace{\gamma_{\mu cd}\Big(\sum_r u^r_d \bar{u}^r_g\Big)\gamma_{\nu gh}\Big(\sum_{s'} u^{s'}_h \bar{u}^{s'}_c\Big)}_{=\mathrm{Tr}(\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_3)}$$

$$\langle|\mathcal{M}_u|^2\rangle = \frac{e^4}{4u^2}\mathrm{Tr}\left[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_4\right]\mathrm{Tr}\left[\gamma_\mu \not{p}_2 \gamma_\nu \not{p}_3\right]$$

We now use the trace identity we used for $\langle|\mathcal{M}_t|^2\rangle$

$$= 8\frac{e^4}{u^2}\left[(p_1\cdot p_2)(p_4\cdot p_3) + (p_1\cdot p_3)(p_4\cdot p_2)\right]$$

$(p_1\cdot p_2) = \frac{1}{2}s = (p_4\cdot p_3)$

$(p_1\cdot p_3) = -\frac{1}{2}t = (p_4\cdot p_2)$

$$\langle|\mathcal{M}_u|^2\rangle = \frac{2e^4}{u^2}(s^2+t^2)$$

Now we calculate the cross term $\langle \mathcal{M}_{tu}\rangle = \frac{1}{4}\sum_{ss'rr'} -2\mathrm{Re}\{\mathcal{M}_t^\dagger \mathcal{M}_u\}$

$$\langle \mathcal{M}_{tu}\rangle = -\frac{1}{2}\sum_{ss'rr'}\mathrm{Re}\left(-\frac{ie^2}{t}(\bar{u}^s\gamma^\mu u^{s'})(\bar{u}^r\gamma_\mu u^{r'})\frac{ie^2}{u}(\bar{u}^{r'}\gamma^\nu u^s)(\bar{u}^{s'}\gamma_\nu u^r)\right)$$

In component form: $\langle \mathcal{M}_{tu}\rangle = -\frac{e^4}{tu2}\sum_{ss'rr'}\mathrm{Re}\{\bar{u}^s_a\gamma^\mu_{ab}u^{s'}_b\bar{u}^r_c\gamma_{\mu cd}u^{r'}_d\bar{u}^{r'}_e\gamma^\nu_{ef}u^s_f\bar{u}^{s'}_g\gamma_{\nu gh}u^r_h\}$

$$\langle \mathcal{M}_{tu}\rangle = -\frac{e^4}{2ut}\mathrm{Re}\Big\{\gamma^\mu_{ab}\Big(\sum_{s'}u^{s'}_b\bar{u}^{s'}_g\Big)\gamma_{\nu gh}\Big(\sum_r u^r_h\bar{u}^r_c\Big)\gamma_{\mu cd}\underbrace{\Big(\sum_{r'}u^{r'}_d\bar{u}^{r'}_e\Big)}_{\not{p}_{4\,de}}\gamma^\nu_{ef}\underbrace{\Big(\sum_s u^s_f\bar{u}^s_a\Big)}_{\not{p}_{1\,fa}}\Big\}$$

$$= -\frac{2e^4}{4ut}\mathrm{Re}\left\{\mathrm{Tr}\left[\gamma^\mu\not{p}_3\gamma_\nu\not{p}_2\gamma_\mu\not{p}_4\gamma^\nu\not{p}_1\right]\right\}$$

In previous exercise we get:

$$-\frac{e^4}{2st}\mathrm{Re}\left\{\mathrm{Tr}\left[\gamma^\mu\not{p}_a\gamma_\nu\not{p}_b\gamma_\mu\not{p}_c\gamma^\nu\not{p}_d\right]\right\} = 16\frac{e^4}{st}\mathrm{Re}\{(p_a\cdot p_c)(p_b\cdot p_d)\}$$

$$\Rightarrow \langle \mathcal{M}_{tu}\rangle = 16\frac{e^4}{ut}\mathrm{Re}\{(p_3\cdot p_4)(p_2\cdot p_1)\} \quad \Big\| \quad (p_3\cdot p_4) = \frac{1}{2}s = (p_2\cdot p_1)$$

$$= 16\frac{e^4}{ut}s^2\cdot\frac{1}{4}$$

$$\langle \mathcal{M}_{tu}\rangle = \frac{4e^4}{ut}s^2 \quad \Rightarrow \quad |\mathcal{M}|^2 = \langle|\mathcal{M}_t|^2\rangle + \langle|\mathcal{M}_u|^2\rangle + \langle \mathcal{M}_{tu}\rangle =$$

$$\Rightarrow |\mathcal{M}|^2 = \frac{2e^4}{t^2}(u^2+s^2) + \frac{2e^4}{u^2}(s^2+t^2) + \frac{4e^4}{ut}s^2$$

$$= 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{2s^2}{ut}\right]$$

We can see that this is same as the Matrix element we get by applying crossing symmetry of $u\to s$, $t\to u$, $s\to t$ to the solution of previous exercise

The cross-section in Centre of Mass (CM) Frame is

$$\left(\frac{d\sigma}{d\cos\theta}\right)_{CM} = -\frac{1}{32\pi s}|\mathcal{M}|^2$$ || ( Equation from previous exercise )
|| Now we put in the value of $|\mathcal{M}|^2$

$$\left(\frac{d\sigma}{d\cos\theta}\right)_{CM} = -\frac{2e^4}{32\pi s}\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{2s^2}{ut}\right]$$ || $\alpha = \frac{e^2}{4\pi} \Rightarrow e^4 = \alpha^2 16\pi^2$

$$\left(\frac{d\sigma}{d\cos\theta}\right)_{CM} = -\frac{\alpha^2\pi}{s}\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{2s^2}{ut}\right]$$

Since the expression is in terms of Lorentz invariant quantities it is valid in any frame:

$$\boxed{\frac{d\sigma}{d\cos\theta} = -\frac{\alpha^2\pi}{s}\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{2s^2}{ut}\right]}$$